

➤ Causes of corneal transparency:

1. Avascular (Q. How does the cornea receive its O₂ supply & nutrients?)

N.B. Vitamin B2 (riboflavin) ↓↓ → corneal vascularization → loss of corneal transparency

2. Unmyelinated nerve endings

3. Regular arrangement of corneal epithelium & collagen fibers

4. Relative corneal dehydration: *via*

(a) Metabolic pump: of Na⁺ from cornea to AH by Na⁺-K⁺ pump

(b) Osmosis: H₂O follows Na⁺ to AH & tear film

N.B. Vitamin A ↓↓ → xerophthalmia (dry eye) → impairment of corneal transparency

4. Sclera

➤ Functions:

1. Protection of inner delicate structures

2. Gives attachment to extraocular ms. (EOM)

Lecture (17): Functions of the different parts of the eye (II)

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5. Crystalline Lens

➤ Structure & Properties:

1. Biconvex transparent structure behind the iris (its post. surface is more convex than the ant. surface during rest)

2. Enclosed in an elastic capsule, but it becomes progressively hard with

advance of age

3. The younger fibers form the cortex, while the older fibers form the nucleus

4. Suspended to the ciliary body by SL

5. Avascular → its metabolism is mainly anaerobic

N.B. Some aerobic metabolism occur & need glutathione (antioxidant)

➤ Functions:

1. One of the refractive media of the eye

2. Accommodation for near vision

3. Protection of retina from UV rays

➤ Causes of lens transparency:

1. Avascular (Q. How does the lens receive its O₂ supply & nutrients?)

2. No nerve supply

3. Lens fibers are uniformly arranged & densely packed

4. Various lens constituents have almost the same RI

➤ Cataract:

*Definition: Degenerative disease of the crystalline lens with partial or complete loss of its transparency

*Types: 1. Nuclear 2. Cortical 3. Post. subcapsular

*Causes:

1- ↓↓ glutathione → ↑↑ O₂ free radicals → ↑↑ permeability to H₂O & ions **“Senile cataract”**

2. Denaturation of lens proteins with deposition of Ca²⁺ salts (*e.g.*, in DM, excessive exposure to UV rays or high temperatures, previous eye surgery or injury & prolonged use of corticosteroids)

*C/P:

1. Blurred vision
2. Poor night or day vision (?)
3. Sensitivity to light & glare
4. Seeing ‘halos’ around lights
5. Fading of colors
6. Double vision in a single eye

*ttt: Surgical removal of the lens (by extracapsular extraction or phacoemulsification) + IOL implantation or placing a powerful convex lens in front of the eyes (*i.e.* glasses)

➤ Near response (Near reflex):

*Stimulus: Seeing a near object

*Receptor: Photoreceptors (rods & cones)

*Afferent: Optic nerve → Optic chiasma → Optic tract → LGB → Optic radiations → Visual cortex (areas 17, 18 & 19) → Descending occipito-tectal tracts →

*Center: Sup. colliculus in MB

*Efferent: EWN on both sides → 3rd CN:

(a) **Parasymp. part**: Preganglionic fibers → ciliary ganglion → postganglionic fibers “short ciliary nerves” → 1. ciliary ms. 2. constrictor pupillae ms.

(b) **Motor part**: motor fibers pass directly → 3. the medial recti

*Response:

1. Accommodation (involuntary)
2. Bilateral miosis (involuntary)
3. Convergence of both eyes (voluntary)

*Significance:

1. Accommodation: to see the near object clear
2. Bilateral miosis: to
 - (a) prevent entry of excess light rays into the eye
 - (b) prevent spherical & chromatic aberrations (see later)
 - (c) increase the depth of the focus (see later)
3. Convergence of both eyes: to prevent diplopia
(How?) (Q. What is the effect of atropine on near reflex?)

6. Choroid

➤ Functions:

1. Vascular: it provides eye with its blood supply

2. Rich in melanin pigment: it converts the interior of eye to a dark chamber (Why?)
3. It gives attachment to ciliary ms.
4. It supports retina & keeps it in place

7. Ciliary Body

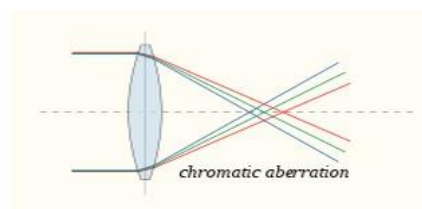
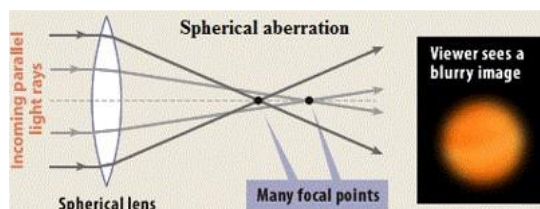
➤ Functions:

1. It contains ciliary processes which secrete AH into the post. chamber
2. It contains ciliary ms. which is responsible for accommodation
3. It gives attachment to SL of lens

8. Iris

➤ Functions:

1. It regulates the amount of light entering the eye through the pupil:
 - (a) Constrictor pupillae ms. → miosis
 - (b) Dilator pupillae ms. → mydriasis
2. It protects the retina from UV rays
3. It prevents spherical & chromatic aberrations (How?)



4. It controls the depth of focus (How?)

➤ Pupillary light reflex (PLR):

*Stimulus: Exposure of one eye to light

*Receptor: Photoreceptors (rods & cones)

*Afferent: Optic nerve → Optic chiasma → Optic tract (no relay in LGB)

*Center: Pretectal nucleus in MB

*Efferent: EWN on both sides (parasymp.) → 3rd CN (preganglionic fibers → ciliary ganglion → postganglionic fibers “short ciliary nerves”)

*Response: Bilateral miosis (cont. of constrictor pupillae ms.):

(a) Direct: in the stimulated eye

(b) Consensual or indirect: in the unstimulated eye

*Significance:

1. Protection: it decreases the amount of light entering the eye
2. Diagnosis of some nervous lesions & diseases
3. Staging of anesthesia (How?)

➤ Argyll-Robertson pupil:

*Cause: Lesion of pretectal nucleus (near the aqueduct of Sylvius) due to neurosyphilis

*Effect: PLR is lost, while AR is intact (*i.e.*, pupil does not constrict on exposure to light, but it constricts with accommodation)

➤ Reverse of Argyll-Robertson pupil:

*Cause: Bilateral damage of occipito-tectal tracts

*Effect: PLR is intact, while AR is lost

➤ **Conditions associated with miosis vs mydriasis:**

Conditions ass. with miosis	Conditions ass. with mydriasis
1. Light exposure (PLR)	1. Sudden withdrawal of light
2. Light adaptation	2. Dark adaptation
3. Near vision	3. Far vision
4. Parasymp. stimulation, <i>e.g.</i> , during sleep	4. Symp. stimulation, <i>e.g.</i> , emotions, pain & stress
5. Lesion of cervical symp. chain (Horner's syndrome)	5. Lesion of 3 rd CN or EWN
6. Drugs: (a) Parasympathomimetics, <i>e.g.</i> , eserine & pilocarpine (b) Morphine poisoning (!) (c) Histamine	6. Drugs: (a) Sympathomimetics, <i>e.g.</i> , adrenaline & phenylephrine (b) Parasympatholytics, <i>e.g.</i> , atropine (c) Cocaine
7. 3 rd stage of general anesthesia	7. 2 nd & 4 th stages of general anesthesia
8. Sudden fall in IOP	8. Increased IOP (glaucoma)
9. Pontine hge (!) & TS of spinal cord above T ₁	9. Hyperthyroidism

9. Retina

➤ **Functional organization:**

Retina is formed of 10 layers (from outside to inside):

1. Pigmented epithelial layer: = Single layer of epithelial cells containing melanin

***Functions:**

i) Absorbs light rays not absorbed by photoreceptors

- ii) Stores vit. A (the precursor of photopigments)
- iii) Phagocytoses the debris of photoreceptors
- iv) Forms the “outer blood-retinal barrier” (?!)
- v) Provides mechanical support to photoreceptors

2. Photoreceptor layer (rods & cones): = Receptors that convert light rays into nerve impulses

3. Outer limiting membrane: = Formed by tight junctions between the cell membranes of photoreceptors & Muller’s cells (not a true BM)

4. Outer nuclear layer: = Contains nuclei of photoreceptors

5. Outer plexiform layer: = Contains synapses between cells in layers 4 & 6

6. Inner nuclear layer: = Contains nuclei of bipolar “1st order neuron”, horizontal & amacrine cells + Muller’s cells “supporting cells”

7. Inner plexiform layer: = Contains synapses between cells in layers 6 & 8

8. Ganglion cell layer: = “2nd order neuron” *Their dendrites receive impulses from bipolar cells *Their axons form optic nerve fibers

9. Optic nerve fibers layer: = Axons of ganglion cells → leave retina through optic disc

10. Inner limiting membrane: = A true BM that forms interface between retina & vitreous humour

➤ Regional variation:

(1) Macula lutea:

- Yellowish spot
- 1 mm in diameter
- Opposite to the post. pole of eye
- 3 mm from the temporal side of optic disc

(2) Fovea centralis:

- Central depression in the macula lutea
- 0.4 mm in diameter
- Has the highest visual acuity, due to:
 - (a) Formed of cones only
 - (b) One to one connection (no convergence)
 - (c) No other layers (light rays fall directly on cones)
 - (d) Highly developed pigmented epithelial layer

(3) Optic disc:

- Area at which the axons of ganglion cells leave the retina & enter the optic nerve
- 1.5 mm in diameter
- 3 mm med. & slightly above the post. pole of eye
- Contains no photoreceptors → “physiological blind spot” in the visual field → ignored normally (as the 2 blind spots are not on the same corresponding points on both retinae)